Status	Finished
Started	Tuesday, 20 October 2026, 10:55 PM
Completed	Tuesday, 20 October 2026, 11:15 PM
Duration	20 mins 31 secs
Grade	**14.00** out of 15.00 (**93.33**%)

Question **1**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm is based on repeatedly dividing the list into halves?

a. Bubble Sort
b. Merge Sort
c. Quick Sort
d. Insertion Sort

The correct answer is: Merge Sort

Question **2**

Complete

Mark 1.00 out of 1.00

What is the primary advantage of the divide-and-conquer approach in sorting algorithms?

a. It simplifies the sorting process by using only one pass
b. It avoids the need for recursion
c. It allows for efficient parallel processing and sorting of data
d. It only works on small datasets

The correct answer is: It allows for efficient parallel processing and sorting of data

Question **3**

Complete

Mark 1.00 out of 1.00

In Merge Sort, what happens after the two halves of the list are sorted?

a. They are compared element by element
b. They are split again into smaller sublists
c. They are combined to form a single sorted list
d. They are discarded

The correct answer is: They are combined to form a single sorted list

Question **4**

Complete

Mark 1.00 out of 1.00

What is the primary benefit of using sorting algorithms in programming?

a. Provides a basis for other algorithms to work efficiently
b. Makes data harder to manage
c. Makes code execution slower
d. Decreases the efficiency of algorithms

The correct answer is: Provides a basis for other algorithms to work efficiently

Question **5**

Complete

Mark 1.00 out of 1.00

Why is sorting important for selection operations?

a. It complicates the selection of items
b. It slows down the process
c. It makes it easier to select items based on their relationship to the rest of the items
d. It makes the data unsorted

The correct answer is: It makes it easier to select items based on their relationship to the rest of the items

Question **6**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm involves comparing elements and swapping adjacent items that are out of order?

- a. Bubble Sort
- b. Binary Search
- c. Linear Search
- d. Merge Sort

The correct answer is: Bubble Sort

Question **7**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm is described as making multiple passes through a list, comparing elements, and swapping adjacent items that are out of order?

- a. Bubble Sort
- b. Quick Sort
- c. Merge Sort
- d. Insertion Sort

The correct answer is: Bubble Sort

Question **8**

Complete

Mark 1.00 out of 1.00

What is a key disadvantage of Bubble Sort compared to more advanced algorithms like Merge Sort?

- a. Bubble Sort cannot handle duplicate elements
- b. Bubble Sort is less efficient for large lists
- c. Bubble Sort does not guarantee sorted order
- d. Bubble Sort is difficult to implement

The correct answer is: Bubble Sort is less efficient for large lists

Question **9**

Complete

Mark 1.00 out of 1.00

What is a characteristic of the merge sort algorithm?

a. It is less efficient than bubble sort
b. It is based on the divide-and-conquer approach
c. It does not require recursion
d. It sorts data using a single pass

The correct answer is: It is based on the divide-and-conquer approach

Question **10**

Complete

Mark 1.00 out of 1.00

_____is putting an element in the appropriate place in a sorted list yields a larger sorted order list.

a. Distribution
b. Selection
c. Insertion
d. Extraction

The correct answer is:
Insertion

Question **11**

Complete

Mark 1.00 out of 1.00

Which built-in Python function is used to sort data?

a. sorted()
b. sort()
c. arrange()
d. order()

The correct answer is: sorted()

Question **12**

Complete

Mark 1.00 out of 1.00

Algorithm design technique used in merge sort algorithm is

- a. Divide and conquer
- b. Dynamic programming
- c. Greedy method
- d. Backtracking

The correct answer is:
Divide and conquer

Question **13**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm is known for its simplicity and sequential comparison of elements?

- a. Quick Sort
- b. Merge Sort
- c. Heap Sort
- d. Bubble Sort

The correct answer is: Bubble Sort

Question **14**

Complete

Mark 1.00 out of 1.00

Two-way merge sort algorithm is used to sort the following elements in ascending order.
200,470,150,80,90,40,400,300,120,70
What is the order of these elements after second pass of the merge sort algorithm?

- a. 40,80,90,150,200,300,400,470,70,120
- b. 80,150,200,470,40,90,300,400,70,120
- c. 200,470,80,150,40,90,300,400,70,120
- d. 40,70,80,90,120,150,200,300,400,470

The correct answer is:
80,150,200,470,40,90,300,400,70,120

Question **15**

Complete

Mark 0.00 out of 1.00

Very slow way of sorting is_____

a. Heap sort

b. Bubble sort

c. Quick sort

d. Insertion sort

The correct answer is:
Insertion sort